

4. New Research Proposal

Ajzenman et al. (2023) research provides one of the first attempts to evaluate a behavioral intervention and its scalable counterpart within the same randomized controlled trial (RCT), comparing human-delivered nudges to rule-based chatbots. Rule-based chatbot performance was not ideal and unable to scale nudges to persuade the participants.

The result raises another question whether more advanced conversational agents, such as large-language-model (LLM) chatbots, can achieve better outcome and serve as a cost-efficient tool to scale nudges which aim at mitigating teacher shortages or other skill gaps? The new research will be based on Ajzenman et al. (2023) research, adding AI driven LLM chatbot into comparison to determine its effectiveness.

4.1 Research Objectives

Compare human guidance, rule-based chatbots, and LLM-based chatbots within a unified experimental framework. The goal is to determine not only whether advanced chatbots can replicate the effectiveness of human guidance, but also to identify the mechanisms, heterogeneous effects, and scalability potential of chatbot produced nudges in shaping students' educational and career decisions. General structure of experiment, please refer to Flow Chart 1 in the attachment

4.2 Experimental Design

4.2.1 Target Group

The experiment targets students enrolled in the national university entrance examination system. Compared to Ajzenman et al. (2023), who restricted the sample to students expressing interest in education or social sciences, the proposed design includes:

Core sample: students indicating interest in education or social sciences (replicating the original design).

Extended sample: a random subsample of students who did not indicate interest in teaching careers.

This broader sampling frame allows assessment of whether conversational AI can “expand the consideration set” rather than only influencing marginal students already primed to consider teaching (Hoxby & Leigh, 2004).

4.2.2 Baseline Survey

One month before application, students complete a short baseline questionnaire to collect basic information. The baseline survey includes: perceived prestige of working as teachers, medical care staff, civil servants etc.; expected wages; beliefs about job stability and working conditions, perceived competence; initial interest in specific majors.

Although the research is targeting mainly at potential future teachers, the baseline survey is not limited within teaching profession. Responses will be recorded and linked to corresponding student file that is collected from central education application systems (subject to consent). The file includes students' preference lists, admissions decisions, and final enrollment information.

4.2.3 Stratified Randomization

Students are randomly assigned to one of five experimental groups using stratified randomization. The stratification is based on several key characteristics, including gender, type of high school (public, subsidized, or private), socioeconomic background, and academic performance (top 30% vs lower 70%). This procedure helps reduce imbalances across groups and improves the precision of the estimated treatment effects, while also facilitating meaningful comparisons between the different experimental arms (Angrist & Pischke, 2010).

4.2.4 Experimental Arms

The experiment is consisted of five different group to establish needed comparison and benchmarks.

Control Group

The control group receives no additional communication beyond standard information provided by the Ministry of Education and university website. This establishes the baseline against which all interventions are compared.

Information-Only Group

Information-only Group sends three static WhatsApp messages during the week before the application deadline. Messages include basic information about teaching salaries and career progression, employment rates and the social importance of educational careers.

No follow-up questions or interactive elements are permitted in this group. This arm provides a lower bound on the expected effectiveness of non-interactive nudges.

Rule-Based Chatbot Group

Following the design in Ajzenman et al. (2023), students interact with a rule-based chatbot through scripted WhatsApp conversations. The interaction follows a predefined decision tree: the bot poses structured questions about students' motivations and responds with fixed messages. Information on application procedures and financial aid is provided only when students explicitly indicate interest in teaching. Although this approach is relatively inexpensive and straightforward to implement, it leaves little room for adjustment once the conversation starts. In particular, the chatbot is unable to respond flexibly to unexpected questions or to address more nuanced emotional concerns expressed by students.

LLM-Based Chatbot Group

Introducing the LLM chatbot makes up the key difference of this experiment. AI powered large-language-models, unlike rule based chatbots, can tailor its responses to better fit the needs of students, provide certain level of emotional values like human advisors. Prior research demonstrates that AI-based conversational systems can increase user engagement and completion rates relative to rule-based bots (Luo et al., 2019).

Key design features include:

1. a standardized initial prompt to ensure a consistent motivational framework
2. ability to probe ambiguous responses (“What makes you unsure about teaching?”)
3. personalized encouragement strategy based on the student's academic performance, socioeconomic background, or expressed values;
4. constraints to ban over-directive messaging, to ensure ethical communications and students' freedom of choice.

Human Advisor Group

Well-trained human advisors (especially professional advisors and recent graduates) provide WhatsApp conversations following a standard script. Advisors receive the same standardized training to ensure treatment fidelity. Certain flexibility is still allowed for conducting natural conversation. The interaction time of Human Advisor groups will be limited to the same as two

chatbot groups to avoid incomparability. This group serves as the benchmark for measuring the effectiveness of chatbot-based nudges.

4.2.5 Intervention Timing and Implementation

The intervention starts **three weeks before the university major selection deadline**, a period when students are actively planning and adjusting their options. Timing follows evidence that nudges are most effective when delivered close to key decision points (Thaler & Sunstein, 2008).

Each intervention proceeds as follows:

Week 1: establish first contact, motivational assessment, and application preparation content.

Week 2: optional follow-up conversation to address students' questions and concerns.

Week 3: reminder conversations about application deadlines, scholarships, requirements.

Interaction logs—which include message count, length, time stamps, session duration, and branching paths—are stored for mechanistic analysis.

4.2.6 Outcomes and Measurement Strategy

Primary Outcomes

1. First-choice selection of an education programme. (Indicates strong interest and priority in the student's application list.)
2. Proportion of education programs in the full choice set. (Measures the intensive margin of consideration.)
3. Application to at least one education program (Captures the extensive margin, i.e., whether students who were not previously considering teaching can be nudged into applying.)

Secondary Outcomes

1. Final enrollment in an education program
2. Belief updating and motivational shifts
3. Short post-treatment survey measuring
4. salary expectations
5. perceived prestige
6. perceived competence
7. perceived stability of teaching careers

These surveys allow identification of the channels through which conversational agents affect choices. While informative, the enrollment outcome must be interpreted cautiously due to general-equilibrium effects in centralized matching (Ajzenman et al., 2023).

4.2.7 Mechanism Analysis

Interaction logs provide data for analysis covering conversation length, frequency of student-initiated questions, sentiment and emotional tone, dropout rates during conversations.

These parameters can help distinguish whether LLMs outperform rule-based bots through more persuasive communication, or through simply increasing exposure time.

Heterogeneous Treatment Effects

Following Ajzenman et al. (2023), the design uses causal forest estimators to explore heterogeneity along four aspects which includes: academic achievement; gender; socioeconomic status; baseline beliefs about teaching profession.

This approach enables the identification of the subgroups that are most responsive to chatbot-based nudges.

4.2.8 Ethical Considerations

Considering the use of LLM-chatbot communication and experiment's impact on students' life path, strong ethical safeguards shall be imposed:

Only under consent can Chatbot provide guidance towards students;

Chatbots are not allowed to pose any pressure on students or present teaching as the only option;

All conversations between participants and advisors/chatbots shall be logged and monitored for safety;

Human advisors or AI models must not collect sensitive information that exceeds students' consent.

4.2.9 Expected Results

Based on results from previous research, following results are expected:

1. Human advisors are likely to produce the strongest effects on students' preferences for education majors, consistent with findings in Ajzenman et al. (2023), due to their ability to provide personalized guidance with adequate empathy.
2. the large-language-model (LLM) chatbot is expected to close much of the gap between humans and technology, producing treatment effects significantly larger than those of the rule-based chatbot due to its better capacity for open-ended dialogue, adaptive motivational framing, and tailored responses

3. the rule-based chatbot is expected to show modest but positive effects, similar in direction but substantially smaller in magnitude.

Overall, the experiment is expected to show that advanced chatbot systems can partially replicate the nudge from human advisors while offering a feasible path toward scaling nudges to reduce skill shortages, and may outperform human advisors in certain aspect.

5. Conclusion

This thesis examined whether chatbots can scale behavioral nudges to address skill shortages. Evidence from Ajzenman et al. (2023) shows that rule-based chatbots fall short of human-delivered interventions, particularly in influencing emotionally charged, high-stakes career decisions like entering the teaching profession. Their limited personalization, low flexibility, and lack of emotional intelligence prevent them from fully replicating the motivational impact of human counselors.

Nevertheless, recent advances in large language models suggest that the limitations of early chatbot systems are not inherent to the medium itself. LLM-powered chatbots exhibit conversational depth, adaptability, and context sensitivity previously available only through human interaction. While simple chatbots cannot resolve skill shortages, AI-enhanced systems—particularly as part of hybrid human-AI models—hold significant promise. Extra precautions need to be taken to avoid ethical risks, as these nudges involves chatbots changing human selection.

In conclusion, advanced chatbots can help scale nudges, but only if they are sufficiently sophisticated, ethically deployed, and integrated into broader human-centered support systems. Skill shortage situation is a challenge that is already here and likely to stay. AI powered chatbot systems are promising tools for solving the problem with continuous development and supervision. Future research should also explore how human advisors and chatbots can be combined to create more effective and scalable guidance systems.